

Geldium AI-Driven Delinquency Prediction Model Plan

Objective: Develop an AI-driven model to identify customers at risk of missing payments so that Geldium can prioritize outreach and intervention strategies.

1. Model Logic and Approach

Selected Model: Logistic Regression (primary model). Alternative model explored: Random Forest.

Predictive Modeling Pipeline

- Load customer and payment history data.
- Handle missing values using median/mode imputation.
- Encode categorical variables using One-Hot Encoding.
- Scale numerical variables using StandardScaler.
- Split data into training and testing datasets (80:20).
- Train a Logistic Regression model with class balancing.
- Generate delinquency probability scores and classify customers as high or low risk.
- Evaluate performance and monitor fairness.

Top 5 Input Features

1. Credit Utilization
2. Income
3. Age
4. Debt-to-Income Ratio
5. Account Tenure

Chosen Model Summary

Logistic Regression was selected because it is highly interpretable, transparent, and suitable for binary classification problems such as delinquency prediction. It enables business users and regulators to understand how customer characteristics influence risk scores.

2. Model Justification

Logistic Regression provides the best balance between accuracy, interpretability, and ease of deployment. Financial institutions require transparent models that support explainable and responsible AI practices. The model is computationally efficient, easy to monitor, and widely accepted in credit risk analysis. Although more complex models may offer slightly better predictive performance, Logistic Regression is better aligned with Geldium's regulatory and operational requirements.

3. Evaluation Strategy

Performance Metrics

- Accuracy – overall prediction correctness.
- Precision – percentage of predicted delinquent customers that are truly delinquent.
- Recall – ability to identify actual delinquent customers.
- F1 Score – balance between precision and recall.
- ROC-AUC – ability to distinguish between delinquent and non-delinquent customers.

Fairness and Bias Checks

- Compare model performance across demographic groups.
- Review false positive and false negative rates across customer segments.
- Examine feature importance to ensure fairness.
- Conduct periodic audits and retraining to mitigate bias and model drift.

Interpretation and Improvement

Since identifying high-risk customers is the main business objective, Recall and F1-score should be

prioritized over Accuracy alone. Future improvements may include additional feature engineering, collecting richer behavioral data, and testing advanced ensemble techniques.

Conclusion

The proposed AI-driven delinquency prediction framework provides Geldium with an explainable, scalable, and responsible solution for identifying high-risk customers and supporting proactive collection strategies.